

Yiran Peng

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Education

University of Edinburgh	Edinburgh, UK	Sept. 2025 – Now
Degree: MSc Cognitive Science		
Xi'an Jiaotong-Liverpool University	Suzhou, China	Sept. 2021 - Jun. 2025
Degree: BEng Computer Science and Technology		
Honor and Award: Second Prize of the Asia-Pacific Cup Mathematical Modeling Competition (2023)		

Publications

[IEEE SCEMS'23] Yiran Peng, R. Gong, J. Zhou, T. Chen, Q. Liu and X. Wang, "Voltage Harmonic Reduction of Cascaded H-bridge Multilevel Inverter Based on SHEPWM by Using Optimized Immune Algorithm," *2023 IEEE 6th Student Conference on Electric Machines and Systems (SCEMS)*, HuZhou, China, 2023, pp. 1-5, doi: [10.1109/SCEMS60579.2023.10379251](https://doi.org/10.1109/SCEMS60579.2023.10379251) **Accepted**

[IEEE UIC'24] R. Cao, J. He, Yiran Peng, Z. Chen and Q. Sun, "VisionPlay: An Interactive Ball Game Using Computer Vision and Augmented-reality Overlay," *2024 IEEE Smart World Congress (SWC)*, Nadi, Fiji, 2024, pp. 1621-1626, doi: [10.1109/SWC62898.2024.00249](https://doi.org/10.1109/SWC62898.2024.00249) **Accepted**

Project Experience

Remote and Real-time manipulation for Unmanned Surface Vehicles (USVs) based on AR May. 2024 – May.2025

Advisor: Xiaohui Zhu, Yong Yue

- Implemented a **gaze-based** interaction system on **HoloLens 2**, collecting **eye-tracking data** (fixation duration, AOIs, gaze latency) to enable intuitive AR-based USV control
- Applied **gaze behavior modeling and intention inference** to distinguish exploratory and selective attention, mitigating the **Midas Touch problem** through **data-driven analysis of user interaction behavior**
- Developed a **Unity-based AR visualization system** with real-time video streaming, supporting **quantitative evaluation of situational awareness and navigation performance**

A Smart Blue Algae Treatment System based on Unmanned Surface Vehicles (USVs)

Jun. - Aug. 2024

Advisor: Xiaohui Zhu, Yong Yue

- Designed and optimized **Venturi jet pump** by installing 3 bearings to reduce friction, through **Computational Fluid Dynamics** (CFD) simulation for best bacterial inhibitor spraying effect
- Implemented an algorithm to realize real-time control of spraying quantity of inhibitor based on parameters including USV speed, cyanobacteria concentration, river depth, and spray width
- Conducted field experiment by spraying inhibitor powder **5g per second** in *Dushu Lake, Suzhou*, declining density of cyanobacteria by approximately **60%**

Vision Play: Interactive Ball Game with Computer Vision and Real-time Project

Feb. - May. 2024

Advisor: Qilei Sun, Quan Zhang, Jie Sun

- Utilized **Python** and **OpenCV** for real-time ball tracking and collision detection, transmitting coordinates between **Python backend and Unity via sockets**
- Designed game environments and rules in Unity using **C#**, including **animations, scene transitions, and particle effects** in both conventional "Whack-a-Mole" and customized "Project Bode"
- Implemented the **Intersection-over-Union** (IoU) method to validate system accuracy, achieving over **75%** accuracy in object detection and tracking achieved under various circumstances
- Conducted a qualitative analysis using **SPSS**, applying the **SUS** (System Usability Scale) and **UES** (User Engagement Scale) for usability analysis. Both interactive games scored above **70**, significantly higher than the PC version. **ANOVA** analysis revealed significant differences in user experience among the three game modes ($F(2,57) = 37.400, p < .001$)

Comparative Study of Optimization Algorithms for SHEPWM in a Cascaded H-bridge Multilevel Inverter

Jun. - Aug. 2023

Advisor: Xucheng Wang, Qinyao Liu

- Integrated the Immune Algorithm (**IA**) with Selected Harmonic Elimination Pulse Width Modulation (**SHEPWM**) using **MATLAB** to solve **nonlinear equations** and obtain continuous valid **switching angles**
- Define the initial population of switching angles, calculate the **affinity function**, and perform **immune operations** (selection, cloning, mutation). Improve SHEPWM switching angle calculation efficiency by checking **boundary conditions**
- Utilized **PLECS** for simulation experiments, comparing Particle Swarm Optimization (**PSO**) and Genetic Algorithm (**GA**). **IA** shows better performance in reducing Total Harmonic Distortion (**THD**) to as low as **2%** an the line voltage Fast Fourier Transform (**FFT**) to nearly **zero** as the cascade number increases

Extracurricular Activities

Student Accommodation Committee, XJTLU Vice-Chair Oct. 2021 - Jun. 2023

- Organized several major events, including the club recruitment events
- Operated the school social media platform; created and published articles and video contents

Skills

Programming language: MATLAB, C#, C++, Java, Python, JavaScript, SQL, Assembly

Tools and Frameworks: Unity, Git, MySQL, OpenCV, Simulink, PIECS, Wireshark